

MCCHORD AFB, WA, USA

WMO: 742060

Lat: 47.150N

Lon: 122.483W

Elev: 98

StdP: 100.15

Time Zone: -8.00 (NAP)

Period: 94-19

WBAN: 24207

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	-6.1	-4.0	-10.4	1.6	-2.7	-7.2	2.1	-2.3	11.5	8.3	10.1	7.6	1.1	180	0.531

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	12.9	30.5	18.7	28.3	18.0	26.4	17.2	19.8	27.8	18.9	26.4	18.0	24.8	3.6	30

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
17.2	12.4	21.4	16.2	11.7	20.6	15.3	11.0	19.7	56.7	27.9	53.6	26.3	50.8	24.8	22.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
9.0	7.9	6.9	-10.1	34.1	2.8	2.1	-12.1	35.7	-13.8	36.9	-15.3	38.2	-17.3	39.7		
			-10.4	21.0	2.8	1.1	-12.4	21.8	-14.0	22.4	-15.6	23.1	-17.6	23.9		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	10.7	4.6	5.1	7.2	9.4	12.8	15.5	18.4	18.5	15.5	10.5	6.7	4.1
	DBStd	5.89	3.50	3.08	2.67	2.68	2.96	2.76	2.55	2.28	2.52	2.71	3.48	3.45
	HDD10.0	771	169	137	94	43	7	0	0	0	1	27	108	184
	HDD18.3	2854	427	369	346	269	172	93	29	26	90	243	350	441
	CDD10.0	1037	2	1	7	24	95	166	262	263	165	42	8	2
	CDD18.3	77	0	0	0	0	1	9	32	31	4	0	0	0
	CDH23.3	1306	0	0	1	12	57	175	497	437	127	2	0	0
	CDH26.7	389	0	0	0	2	9	57	169	129	24	0	0	0
(14) Wind	WSAvg	2.7	2.5	2.7	3.1	3.1	2.9	2.9	2.7	2.5	2.4	2.5	2.7	2.7
(15) Precipitation	PrecAvg	1062	150	102	114	86	62	47	18	26	44	102	165	147
	PrecMax	1443	312	218	228	188	109	100	48	94	197	244	365	293
	PrecMin	793	60	4	39	25	7	8	1	1	3	19	42	31
	PrecStd	169	58	51	44	41	28	24	12	23	40	50	78	65
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	14.3	15.2	20.3	24.9	27.9	31.6	33.8	32.7	29.4	22.5	17.2	14.0
		MCWB	11.5	10.0	12.4	15.0	16.8	18.5	19.7	19.4	17.7	14.5	13.7	12.2
	2%	DB	12.5	13.0	16.7	20.3	24.9	27.9	30.8	29.9	26.9	19.3	14.8	11.9
		MCWB	10.7	9.7	11.1	12.8	15.7	17.3	18.8	19.0	17.0	14.0	12.6	10.4
	5%	DB	11.1	11.7	14.1	17.5	22.3	24.9	28.2	27.8	24.5	17.4	13.1	10.2
		MCWB	9.7	9.1	10.1	11.3	14.5	16.2	18.2	18.0	16.4	13.3	11.3	9.1
	10%	DB	9.8	10.2	12.5	15.2	19.8	22.4	26.0	25.8	22.3	15.9	12.1	9.0
		MCWB	8.7	8.2	9.1	10.2	13.6	15.3	17.3	17.4	15.5	12.6	10.6	7.9
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	13.0	12.6	13.7	15.6	18.3	20.3	21.3	20.9	19.8	17.0	14.9	12.7
		MCDB	14.2	13.6	17.9	22.7	24.8	28.8	31.4	29.1	25.1	19.3	16.1	13.9
	2%	WB	11.3	10.9	12.1	13.6	16.6	18.1	19.7	19.8	18.2	15.3	13.1	10.7
		MCDB	11.9	12.0	15.3	18.7	22.9	25.5	28.5	27.6	23.7	17.5	14.3	11.4
	5%	WB	9.9	9.6	10.9	12.1	15.3	16.8	18.8	18.8	17.1	14.1	11.9	9.3
		MCDB	10.6	10.7	13.2	16.0	20.9	23.6	26.9	26.2	22.5	16.2	12.9	10.0
	10%	WB	8.6	8.4	9.8	11.0	14.1	15.7	17.8	17.9	16.3	13.1	10.7	8.0
		MCDB	9.4	9.9	12.1	14.3	19.0	21.5	25.1	24.5	21.1	15.1	11.6	8.6
(35) Mean Daily Temperature Range	5% DB	MDBR	7.1	8.6	9.6	10.5	11.0	11.3	12.9	13.0	12.3	10.0	8.3	6.7
		MCDBR	8.4	11.1	13.3	15.6	16.9	17.1	18.2	17.8	17.4	13.1	9.2	7.4
		MCWBR	6.3	7.5	8.2	8.6	8.5	7.8	7.7	7.8	8.7	8.2	6.7	5.9
	5% WB	MCDBR	7.0	8.2	11.2	12.9	14.3	14.8	15.9	15.5	14.1	10.3	7.7	6.4
		MCWBR	5.9	6.2	7.5	7.5	7.6	7.1	7.1	7.1	7.4	7.1	6.0	5.5
(40) Clear-Sky Solar Irradiance	taub		0.325	0.316	0.327	0.356	0.376	0.366	0.350	0.356	0.344	0.329	0.323	0.332
	taud		2.478	2.536	2.525	2.416	2.381	2.433	2.498	2.487	2.517	2.561	2.519	2.391
	Ebn at Noon		762	862	902	899	889	897	908	889	871	832	759	694
	Edh at Noon		66	76	88	108	116	111	102	100	88	73	63	64
(44) All-Sky Solar Radiation	RadAvg		1.02	1.88	2.80	4.10	5.19	5.68	6.34	5.43	3.99	2.26	1.20	0.84
	RadStd		0.15	0.26	0.34	0.39	0.44	0.58	0.48	0.38	0.37	0.25	0.16	0.12

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	+1.00	+1.30	N/A	N/A	N/A	N/A
(47) Regional (57 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page